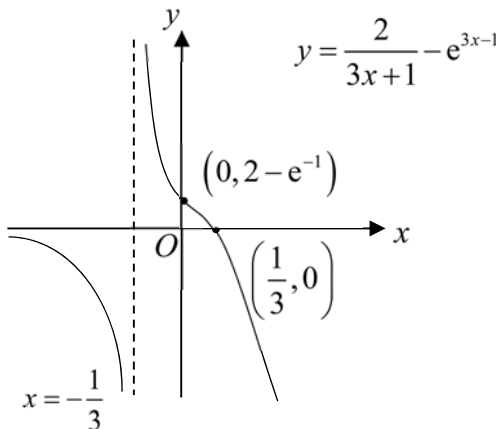


2018 CJC JC2 H1 Prelim Solution

1	
	Since $kx^2 - 4x + k < 0$, graph is “n-shape” and there are no real roots. Hence, $b^2 - 4ac < 0$ and $k < 0$ $(-4)^2 - 4(k)(k) < 0$ $16 - 4k^2 < 0$ $4 - k^2 < 0$ $(2+k)(2-k) < 0$ $k < -2$ or $k > 2$ (reject Q $k < 0$) Therefore, $k < -2$.

2	
	Substituting $u = x^2$, $2u^2 + u - 1 = 0$ $(u+1)(2u-1) = 0$ $u = -1 \text{ or } u = \frac{1}{2}$ $\Rightarrow x^2 = -1 \text{ (reject Q } x^2 \geq 0) \text{ or } x^2 = \frac{1}{2}$ $x^2 = \frac{1}{2}$ $x = \pm \frac{1}{\sqrt{2}}$

3	
	(i) When $x = \frac{e-1}{2}$, $k = \frac{2\left(\frac{e-1}{2}\right)}{e-1} = 1$
	(ii) $y = \ln(2x+1)$ $e^y = 2x+1$ $x = \frac{e^y - 1}{2}$ $\int_0^1 \frac{e^y - 1}{2} dy$ $= \frac{1}{2} \int_0^1 e^y - 1 dy$ $= \frac{1}{2} [e^y - y]_0^1$ $= \frac{1}{2} (e - 1 - 1)$ $= \frac{1}{2} (e - 2)$
	(iii) Area $= \frac{1}{2}(1)\left(\frac{e-1}{2}\right) - \frac{1}{2}(e-2)$ $= \frac{1}{4}(3-e)$

4	
	(i)  $y = \frac{2}{3x+1} - e^{3x-1}$ $(0, 2 - e^{-1})$ $\left(\frac{1}{3}, 0\right)$ $x = -\frac{1}{3}$
	(ii) $\frac{dy}{dx} = -2(3)(3x+1)^{-2} - 3e^{3x-1}$ When $x = \frac{1}{3}$, $\frac{dy}{dx} = -6(2)^{-2} - 3e^{\frac{3}{3}-1} = -\frac{9}{2}$ and $y = 1 - e^{\frac{3}{3}-1} = 0$ Equation of tangent: $y - 0 = -\frac{9}{2}\left(x - \frac{1}{3}\right)$ $\frac{9}{2}x + y - \frac{3}{2} = 0$ $9x + 2y - 3 = 0$
	(iii) At y-axis, $x = 0$, $y = \frac{3}{2}$. $\therefore A\left(0, \frac{3}{2}\right)$ At x-axis, $y = 0$, $x = \frac{1}{3}$. $\therefore B\left(\frac{1}{3}, 0\right)$

		Hence, $AB = \sqrt{\left(\frac{1}{3}\right)^2 + \left(\frac{3}{2}\right)^2}$ $= \sqrt{\frac{85}{36}}$ $= 1.5366$ $\approx 1.54 \text{ (to 3 s.f.)}$
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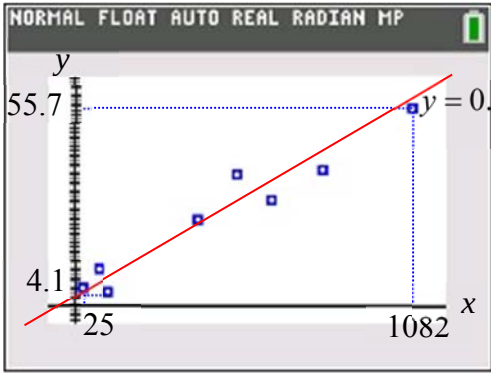
5													
	(i) By Pythagoras' theorem, $\left(30\sqrt{3}\right)^2 = R^2 + H^2$ $R^2 = 2700 - H^2$ By similar triangles, truncated part of the cone has dimensions $\frac{1}{3}R$ and $\frac{1}{3}H$. $V = \frac{1}{3}\pi R^2 H - \frac{1}{3}\pi \left(\frac{1}{3}R\right)^2 \left(\frac{1}{3}H\right)$ $= \frac{1}{3}\pi R^2 H - \frac{1}{81}\pi R^2 H$ $= \frac{26}{81}\pi R^2 H$ $= \frac{26}{81}\pi (2700 - H^2) H$ $= \frac{26}{81}\pi (2700H - H^3)$												
	(ii) At maximum, $\frac{dV}{dH} = 0$ $\frac{26}{81}\pi (2700 - 3H^2) = 0$ $2700 - 3H^2 = 0$ $H^2 = 900$ $H = 30 \text{ Q } H > 0$ <table border="1"><tr><td>H</td><td>30^-</td><td>30</td><td>30^+</td></tr><tr><td>$\frac{dV}{dH}$</td><td>> 0</td><td>0</td><td>< 0</td></tr><tr><td></td><td>$\nearrow$</td><td>---</td><td>$\searrow$</td></tr></table> Alternative:	H	30^-	30	30^+	$\frac{dV}{dH}$	> 0	0	< 0		$\nearrow$	---	$\searrow$
H	30^-	30	30^+										
$\frac{dV}{dH}$	> 0	0	< 0										
	$\nearrow$	---	$\searrow$										

		$\frac{d^2V}{dh^2} = \frac{26}{81}\pi(-6H)$ $< 0 \text{ when } H = 30$ Therefore, V is a maximum when $H = 30$. $V = \frac{26}{81}\pi(2700(30) - (30)^3)$ $= \frac{52000\pi}{3}$
	(iii)	Let x, y and z represent the unit costs of manufacturing waster paper baskets of economy, standard and premium materials in dollars. $200x + 170y + 50z = 540$ $300x + 200y + 60z = 687$ $350x + 300y + 70z = 901.5$ Using G.C. $x = 0.75$ $y = 1.5$ $z = 2.7$ Cost in 2018: $100(\$0.75) + 70(\$1.50) + 50(\$2.70) = \315

6	
	(i) A random sample is a sample where all customers have equal chance of being selected and the selection is done independently.
	(ii) Whether or not a customer uses cashless payment is independent of other customers using cashless payment. <u>OR</u> The probability of customers using cashless payment remains constant at 0.2.
	(iii) Let C be the random variable denoting the number of customers who use cashless payment, out of 12 customers. $\therefore C \sim B(12, 0.2)$ $P(C \geq 3) = 1 - P(C \leq 2)$ $= 0.4416542515$ $\approx 0.442 \text{ (to 3 s.f.)}$
	(iv) Let Y be the random variable denoting the number of groups with at least 3 customers who uses cashless payment, out of 8 groups. $\therefore Y \sim B(8, P(C \geq 3))$ $\Rightarrow Y \sim B(8, 0.4416542515)$ $P(Y = 5) = 0.1637984338$ $\approx 0.164 \text{ (to 3 s.f.)}$

7	
	(i) No. of arrangements = $11! = 39916800$
	(ii) No. of arrangements = $8! \times (2!)^3$ = 322560
	(iii) No. of arrangements = $6! \times 5!$ = 86400
	(iv) <u>Man on extreme left</u> No. of arrangements = $6 \times 10! = 21772800$ <u>Woman on extreme right</u> No. of arrangements = $5 \times 10! = 18144000$ <u>Man on extreme left AND woman on extreme right</u> No. of arrangements = $6 \times 5 \times 9! = 10886400$ Total no. of arrangements = $21772800 + 18144000 - 10886400$ = 29030400 <u>Alternative:</u> <i>Man on extreme left and woman NOT on extreme right</i> No. of arrangements = $6 \times 5 \times 9! = 10886400$ <i>Woman on extreme right and man NOT on extreme left</i> No. of arrangements = $5 \times 4 \times 9! = 7257600$ <i>Man on extreme left AND woman on extreme right</i> No. of arrangements = $6 \times 5 \times 9! = 10886400$

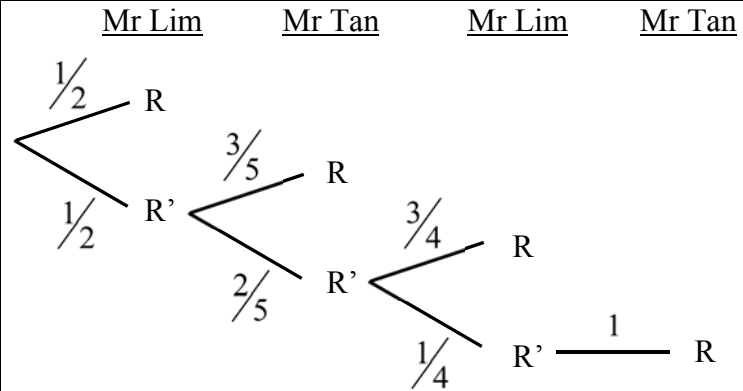
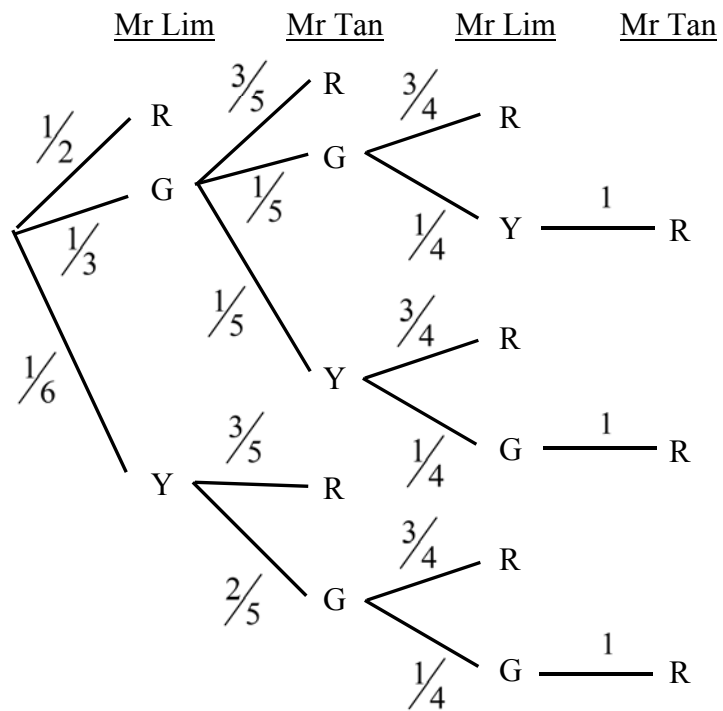
		Total no. of arrangements $= 10886400 + 7257600 + 10886400$ $= 29030400$
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8	
	(i) 
	(ii) Using GC, $r = 0.970220531 \approx 0.970$ (3 s.f.). As the value of r is close to 1, this suggests that there is a strong, positive, linear relationship between the number of international visitor arrivals and GDP per capita. As the number of international visitor arrivals increases, GDP per capita increases proportionally.
	(iii) Using GC, regression line of x on y is $x = 20.04745783y - 61.47085271 \approx 20.0 - 61.5$ (to 3 s.f.) Making y the subject, $y = \frac{x + 61.47085271}{20.04745783} = 0.0498816363x + 3.066266717$ $\approx 0.0499x + 3.07$ (3 s.f.)
	(iv) $y = \frac{45100}{1000} = 45.1$ When $y = 45.1$, $x = 20.04745783(45.1) - 61.47085271$ $= 842.6694954$ $= 843$ (to 3 s.f.)
	The estimated number of visitor arrivals from Canada is 843000. This estimate is reliable as $r = 0.970$ is close to 1 and $y = 45.1$ is within the data range of $4.1 \leq y \leq 55.7$.

9	
	(i) Let K be the random variable denoting the mass, in grams, of a randomly selected Kickers chocolate. $\therefore K \sim N(53, 2.8^2)$ $P(K_1 > 50) \times P(K_2 > 50) = 0.7361838758$ ≈ 0.736 (to 3 s.f.)
	(ii) Let T be the random variable denoting the mass, in grams, of a randomly selected empty tin. $\therefore T \sim N(15, 0.4^2)$ $K_1 + K_2 + K_3 + K_4 + K_5 + T \sim N(5 \times 53 + 15, 5 \times 2.8^2 + 0.4^2)$ $K_1 + K_2 + K_3 + K_4 + K_5 + T \sim N(280, 39.36)$ $P(K_1 + K_2 + K_3 + K_4 + K_5 + T < 275) = 0.2127339029$ ≈ 0.213 (to 3 s.f.)
	(iii) Let V be the random variable denoting the mass, in grams, of a randomly selected Venus chocolate. $\therefore V \sim N(35, \sigma^2)$ $P(V > 34) = 0.85$ $P(V < 34) = 0.15$ $P\left(Z < \frac{34 - 35}{\sigma}\right) = 0.15$ $P\left(Z < \frac{-1}{\sigma}\right) = 0.15$ $\frac{-1}{\sigma} = -1.03643338$ $\sigma = 0.9648473501 \approx 0.9648$ (to 4 d.p.)
	(iv) $2K \sim N(2 \times 53, 2^2 \times 2.8^2)$ $2K \sim N(106, 31.36)$ $3V \sim N(3 \times 35, 3^2 \times 0.9648473501^2)$ $3V \sim N(105, 8.378373681)$

		$2K - 3V \sim N(106 - 105, 31.36 + 8.378373681)$ $2K - 3V \sim N(1, 39.73837368)$ $P(-10 < 2K - 3V < 10) = 0.8828159632 \approx 0.883 \text{ (to 3 s.f.)}$
		The mass of the Kickers chocolates is independent of the mass of the Venus chocolates.

(i)

**Alternative:**

	(ii) $P(\text{Mr Tan wins the game})$ $= \frac{1}{2} \times \frac{3}{5} + \frac{1}{2} \times \frac{2}{5} \times \frac{1}{4} \times 1$ $= 0.35$ <u>Alternative:</u> $P(\text{Mr Tan wins the game})$ $= \frac{1}{3} \times \frac{3}{5} + \frac{1}{6} \times \frac{3}{5} + \frac{1}{3} \times \frac{1}{5} \times \frac{1}{4} \times 1 + \frac{1}{3} \times \frac{1}{5} \times \frac{1}{4} \times 1 + \frac{1}{6} \times \frac{2}{5} \times \frac{1}{4} \times 1$ $= 0.35$
	(iii) $P(\text{Mr Lim wins game on his second turn} \mid \text{Mr Lim wins the game})$ $= \frac{P(\text{Mr Lim wins the game on his second turn})}{P(\text{Mr Lim wins the game})}$ $= \frac{\frac{1}{2} \times \frac{2}{5} \times \frac{3}{4}}{1 - 0.35}$ $= \frac{0.15}{0.65} = \frac{3}{13}$ <u>Alternative:</u> $P(\text{Mr Lim wins game on his second turn} \mid \text{Mr Lim wins the game})$ $= \frac{P(\text{Mr Lim wins the game on his second turn})}{P(\text{Mr Lim wins the game})}$ $= \frac{\frac{1}{3} \times \frac{1}{5} \times \frac{3}{4} + \frac{1}{3} \times \frac{1}{5} \times \frac{3}{4} + \frac{1}{6} \times \frac{2}{5} \times \frac{3}{4}}{1 - 0.35}$ $= \frac{0.15}{0.65} = \frac{3}{13}$
	(iv) $P(\text{Mr Lim has two turns}) = \frac{1}{2} \times \frac{2}{5} = 0.2$

		$P(\text{Mr Lim wins the game}) = 0.65$ $P(\text{Mr Lim has two turns and wins the game}) = 0.15$ Since $0.2 \times 0.65 = 0.13 \neq 0.15$, the two events are not independent.
	(v)	Let X be the random variable denoting the number of games Mr Tan wins, out of 5 games. $\therefore X \sim B(5, 0.35)$ $P(X \leq 2) = 0.764830625 \approx 0.765$ (to 3 s.f.)

11	
	(i) $\hat{\mu} = \frac{286800}{60} + 20000 = 24780$ $s^2 = \frac{1}{59} \left[1429904000 - \frac{286800^2}{60} \right] = 1000000$
	(ii) Let X be the random variable denoting the amount of annual carbon dioxide emission, in tonnes, of a randomly selected company. $H_0 : \mu = 25000$ tonnes $H_1 : \mu < 25000$ tonnes Under H_0, $\mu = 25000$ tonnes, and since $n = 60$ is large enough, by Central Limit Theorem, $\bar{X} \sim N\left(25000, \frac{1000000}{60}\right)$ approximately. Test statistic, $z = \frac{24780 - 25000}{\sqrt{\frac{1000000}{60}}} = -1.704112672$ $p\text{-value} = P(Z < -1.704112672)$ $= 0.0441799877$ Method 1: At 5% significance level, reject H_0 if $p\text{-value} < 0.05$. Since $p\text{-value} = 0.0441799877 < 0.05$, reject H_0. Method 2: At 5% significance level, reject H_0 if $z < -1.644853626$. Since $z = -1.704112672 < -1.644853626$, reject H_0. Conclude that there is sufficient evidence at the 5% significance level that the population mean amount of annual carbon dioxide emission is less than 25000 tonnes.
	(iii) An unbiased estimate is the value of an estimator for the population mean annual carbon dioxide emission where the expected value of the estimator is equal to the population mean annual carbon dioxide emission, i.e. $E(\theta) = \mu$ where θ is the estimator. 5% significance level means that there is a probability of 0.05 of claiming that the population mean amount of annual carbon dioxide emission is less than 25000 tonnes, when it is actually 25000 tonnes.
	(iv) $H_0 : \mu = 25000$ tonnes $H_1 : \mu \neq 25000$ tonnes

Under H_0 , $\mu = 25000$ tonnes, and since $n = 60$ is large enough, by Central Limit Theorem, $\bar{X} \sim N\left(25000, \frac{2000^2}{60}\right)$ approximately.

$$\text{Test statistic, } z = \frac{k - 25000}{\frac{2000}{\sqrt{60}}}$$

At 1% significance level, reject H_0 if $z < -2.575829303$ or $z > 2.575829303$.

Since H_0 is not rejected,

$$-2.575829303 < \frac{k - 25000}{\frac{2000}{\sqrt{60}}} < 2.575829303$$

$$-2.575829303 \left(\frac{2000}{\sqrt{60}} \right) < k - 25000 < 2.575829303 \left(\frac{2000}{\sqrt{60}} \right)$$

$$25000 - 2.575829303 \left(\frac{2000}{\sqrt{60}} \right) < k < 25000 + 2.575829303 \left(\frac{2000}{\sqrt{60}} \right)$$

$$24334.92373 < k < 25665.07627$$

$$\Rightarrow \{k \in \mathbb{I} : 24335 \leq k \leq 25665\}$$